Topic: Step functions with initial value problems

Question: Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 2$.

$$y'' - 3y' + 2y = e^{3t} - \sin(t-2)u(t-2)$$

Answer choices:

A $$Y(s) = \frac{1}{(s-3)(s-2)(s-1)} - \frac{e^{-2s}}{(s^2+1)(s-2)(s-1)} + \frac{2}{(s-2)(s-1)}$$

B $$Y(s) = \frac{1}{(s-3)(s-2)(s-1)} - \frac{1}{(s^2+1)(s-2)(s-1)} + \frac{2}{(s-2)(s-1)}$$

C $$Y(s) = \frac{1}{(s-3)} - \frac{e^{-2s}}{(s^2+1)} + \frac{2}{(s-2)(s-1)}$$

D $$Y(s) = \frac{1}{(s-3)(s-2)(s-1)} - \frac{e^{-2s}}{(s^2+1)(s-2)(s-1)}$$

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Solution: A

Using the Laplace transform formula,

$$\mathscr{L}(u_c(t)g(t-c)) = e^{-cs}\mathscr{L}(g(t))$$

the transformed equation will be

$$\mathscr{L}(y'') - 3\mathscr{L}(y') + 2\mathscr{L}(y) = \mathscr{L}(e^{3t}) - \mathscr{L}(\sin(t-2)u(t-2))$$

$$s^2Y(s) - sy(0) - y'(0) - 3(sY(s) - y(0)) + 2Y(s) = \frac{1}{s-3} - e^{-2s}\left(\frac{1}{s^2+1}\right)$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 2$, then solve for $Y(s)$.

$$s^2Y(s) - s(0) - 2 - 3(sY(s) - 0) + 2Y(s) = \frac{1}{s-3} - e^{-2s}\left(\frac{1}{s^2+1}\right)$$

$$s^2Y(s) - 2 - 3sY(s) + 2Y(s) = \frac{1}{s-3} - e^{-2s}\left(\frac{1}{s^2+1}\right)$$

$$(s^2 - 3s + 2)Y(s) = \frac{1}{s-3} - e^{-2s}\left(\frac{1}{s^2+1}\right) + 2$$

$$Y(s) = \frac{1}{(s-3)(s^2-3s+2)} - \frac{e^{-2s}}{(s^2+1)(s^2-3s+2)} + \frac{2}{(s^2-3s+2)}$$

$$Y(s) = \frac{1}{(s-3)(s-2)(s-1)} - \frac{e^{-2s}}{(s^2+1)(s-2)(s-1)} + \frac{2}{(s-2)(s-1)}$$

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Topic: Step functions with initial value problems

Question: Solve the initial value problem, given $y(0) = 0$ and $y'(0) = 1$.

$$y'' = 2u(t-3)$$

Answer choices:

A $y(t) = t^2 + t$

B $y(t) = t^2 u(t-3) + t$

C $y(t) = 2(t-3)^2 u(t-3) + t$

D $y(t) = (t-3)^2 u(t-3) + t$

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Solution: D

Using the Laplace transform formula,

$$\mathscr{L}(u_c(t)) = \frac{e^{-cs}}{s}$$

the transformed equation will be

$$\mathscr{L}(y'') = 2\mathscr{L}(u(t-3))$$

$$s^2Y(s) - sy(0) - y'(0) = e^{-3s}\left(\frac{2}{s}\right)$$

Substitute the initial conditions $y(0) = 0$ and $y'(0) = 1$, then solve for $Y(s)$.

$$s^2Y(s) - s(0) - 1 = e^{-3s}\left(\frac{2}{s}\right)$$

$$s^2Y(s) = e^{-3s}\left(\frac{2}{s}\right) + 1$$

$$Y(s) = e^{-3s}\left(\frac{2}{s^3}\right) + \frac{1}{s^2}$$

$$Y(s) = e^{-3s}\left(\frac{2}{s^3}\right) + \frac{1}{s^2}$$

Then with the inverse transform formula $\mathscr{L}^{-1}(e^{-cs}F(s)) = f(t-c)u(t-c)$, the inverse transform is

$$y(t) = (t-3)^2u(t-3) + t$$

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Topic: Step functions with initial value problems

Question: Find $Y(s)$, given $y(0) = -1$ and $y'(0) = 1$.

$$y'' + 6y = e^{2t}u(t-7)$$

Answer choices:

A $$Y(s) = e^{-7s}\left(\frac{1}{(s-2)(s^2+6)}\right) + \frac{1}{s^2+6} - \frac{s}{s^2+6}$$

B $$Y(s) = e^{14-7s}\left(\frac{1}{(s-2)(s^2+6)}\right) + \frac{1}{s^2+6} - \frac{s}{s^2+6}$$

C $$Y(s) = e^{14-7s}\left(\frac{1}{(s-2)(s^2+6)}\right)$$

D $$Y(s) = e^{-7s}\left(\frac{1}{(s-2)(s^2+6)}\right)$$

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Solution: B

Remember that each function must be shifted by a proper amount. So, getting things set up for the proper shifts gives us,

$$g(t) = e^{2t}u(t-7)$$

$$g(t+c) = e^{2(t+7)}$$

$$g(t+c) = e^{14}e^{2t}$$

and

$$\mathscr{L}(u_c(t)g(t)) = e^{-cs}\mathscr{L}(g(t+c))$$

$$\mathscr{L}(u_7(t)g(t)) = e^{-7s}\mathscr{L}(e^{14}e^{2t})$$

$$\mathscr{L}(u_7(t)g(t)) = e^{-7s}e^{14}\mathscr{L}(e^{2t})$$

$$\mathscr{L}(u_7(t)g(t)) = e^{14-7s}\mathscr{L}(e^{2t})$$

Then the transformed equation will be

$$\mathscr{L}(y'') + 6\mathscr{L}(y) = \mathscr{L}(e^{2t}u(t-7))$$

$$s^2Y(s) - sy(0) - y'(0) + 6Y(s) = e^{14-7s}\left(\frac{1}{s-2}\right)$$

Substitute the initial conditions $y(0) = -1$ and $y'(0) = 1$, then solve for $Y(s)$.

$$s^2Y(s) - s(-1) - 1 + 6Y(s) = e^{14-7s}\left(\frac{1}{s-2}\right)$$

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$$s^2Y(s) + s - 1 + 6Y(s) = e^{14-7s}\left(\frac{1}{s-2}\right)$$

$$(s^2 + 6)Y(s) = e^{14-7s}\left(\frac{1}{s-2}\right) + 1 - s$$

$$Y(s) = e^{14-7s}\left(\frac{1}{(s-2)(s^2+6)}\right) + \frac{1}{s^2+6} - \frac{s}{s^2+6}$$

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